

# Secret Codes & Files

Post-Assessment 2 · 9 questions · 30 points

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Score: / 30

This is a **post-assessment**. It shows what you learned in this unit. Try every question, and check your answers if you have time.

Work on your own, without a computer. You have about **40 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

## 1. Letters are numbers

3 points

Use these facts: `ord("A")` is 65 and `ord("a")` is 97. Letters are numbered in alphabet order.

| Expression                     | Result |
|--------------------------------|--------|
| <code>ord("E")</code>          |        |
| <code>chr(66)</code>           |        |
| <code>chr(ord("x") - 3)</code> |        |

## 2. Around the alphabet

3 points

(a) What is `30 % 26`?

(1 pt)

\_\_\_\_\_

(b) What is `(24 + 4) % 26`?

(1 pt)

\_\_\_\_\_

(c) Number the letters `a = 0`, `b = 1`, ... `z = 25`. Which letter do you get if you shift `x` forward by 5 and wrap around?

\_\_\_\_\_

### 3. Secret Agent Cipher Lab

4 points

```
def shift_letter(ch, shift):
    pos = ord(ch) - ord("a")
    new_pos = (pos + shift) % 26
    return chr(new_pos + ord("a"))

def encrypt(text, shift):
    result = ""
    for ch in text:
        if ch == " ":
            result = result + ch
        else:
            result = result + shift_letter(ch, shift)
    return result
```

(a) What does `encrypt("cab yz", 3)` return? (2 pts)

(b) What does `encrypt("fde bc", -3)` return? (1 pt)

(c) A message was encrypted with a shift of 7. What shift would you pass to `encrypt` to decrypt it? (1 pt)

### 4. Crack the code

3 points

Use the `encrypt` function from Question 3. You intercepted a secret word but don't know the shift.

```
code = "tmdde"
for guess in range(_____):
    print(guess, encrypt(code, _____))
```

(a) Fill in the blanks so the loop tries every shift from 1 to 25 and decrypts with it. (2 pts)

(b) Which shift reveals a real English word, and what is the word? (1 pt)

## 5. Color Reader

4 points

FILE: COLORS.TXT

```
red
green
blue
yellow
```

```
with open("colors.txt") as f:
    lines = f.readlines()
print(len(lines))
print(lines[-1].strip().title())

count = 0
with open("colors.txt") as f:
    for line in f:
        if "l" in line:
            count = count + 1
print(count)
```

(a) Write exactly what the program prints.

(3 pts)

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(b) What does `f.readlines()` give you? Circle one.

(1 pt)

- ☐ (A) One string holding the whole file      ☐ (B) A list of strings, one per line  
☐ (C) Only the last line      ☐ (D) The number of lines

## 6. Where is the file?

2 points

FOLDER

```
KidCode/
  data/
    scores.txt
  high_scores.py
```

(a) `high_scores.py` runs from the `KidCode` folder. Which line opens `scores.txt`? Circle one. (1 pt)

- ☐ (A) `open("data.scores")`      ☐ (B) `open("scores.txt")`  
☐ (C) `open("data/scores.txt")`      ☐ (D) `open("KidCode/scores")`

(b) What error would `open("scores.txt")` cause?

(1 pt)

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## 7. High Score Keeper

4 points

```
with open("scores.txt", "w") as f:
    f.write("50\n")
with open("scores.txt", "w") as f:
    f.write("80\n")
with open("scores.txt", "a") as f:
    f.write("95\n")
```

(a) What is inside `scores.txt` after the program runs?

(2 pts)

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(b) If the **second** block used "a" instead of "w", what would `scores.txt` contain?

(1 pt)

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(c) If none of the `write` calls had `\n`, what would `scores.txt` contain?

(1 pt)

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## 8. Biggest in a file

4 points

`scores.txt` has one whole number on each line. Write a function `max_from_file(filename)` that opens the file and **returns** the biggest number.

FILE: SCORES.TXT

```
12
45
7
```

For this file, `max_from_file("scores.txt")` returns 45.

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## 9. Save a score

3 points

Write a function `save_score(name, score)` that adds a line like `Ana,95` at the end of `scores.txt`, without erasing what is already there.

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**END OF ASSESSMENT**